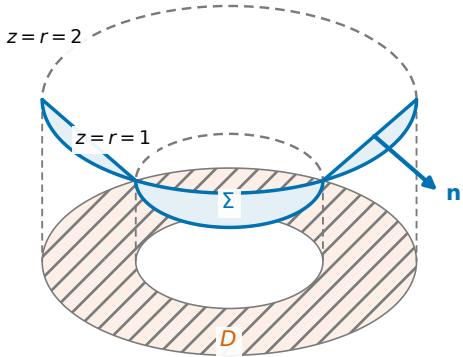
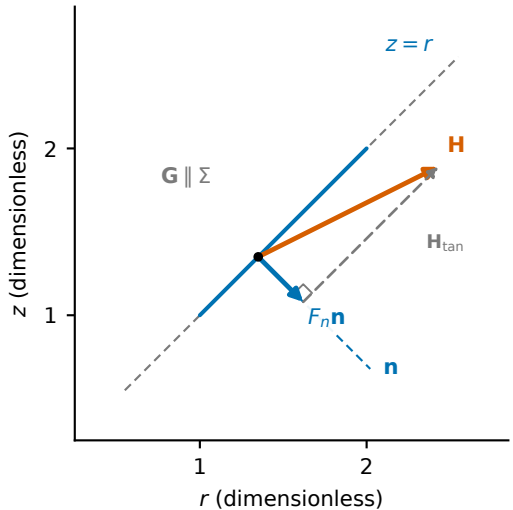
A



$\mathbf{n} = \frac{1}{\sqrt{2}}(x/r,\ y/r,\ -1)$

B



$F_n = r/\sqrt{2}\,, \quad \mathrm{d}S = \sqrt{2}\,\mathrm{d}A_{xy}$